ITI0402 Loogika, sügis 2026
Harjutus 5
Lahendused

1.

$$\dfrac{\dfrac{\dfrac{\overset{+1}{(p \supset q) \wedge (q \supset r)}}{q \supset r}\wedge\mathcal{E} \quad \dfrac{\dfrac{\overset{+1}{(p \supset q) \wedge (q \supset r)}}{p \supset q}\wedge\mathcal{E} \quad \overset{+2}{p}}{q}\supset\mathcal{E}}{\dfrac{r}{p \supset r}\supset\mathcal{I},-2}}{(p \supset q) \wedge (q \supset r) \supset (p \supset r)}\supset\mathcal{I},-1$$

$$\dfrac{\dfrac{\dfrac{\overset{+2}{\neg q} \quad \dfrac{\overset{+1}{p \supset q} \quad \overset{+3}{p}}{q}\supset\mathcal{E}}{\bot}\neg\mathcal{E}}{\neg q \supset \neg p}\supset\mathcal{I},-2}{(p \supset q) \supset (\neg q \supset \neg p)}\supset\mathcal{I},-1$$

$$\dfrac{\dfrac{\dfrac{\overset{+1}{q \wedge r}}{q}\wedge\mathcal{E}}{p \supset q}\supset\mathcal{I},-2}{q \wedge r \supset (p \supset q)}\supset\mathcal{I},-1$$

$$\dfrac{\dfrac{\overset{+1}{p \vee q \supset r} \quad \dfrac{\dfrac{\overset{+2}{q \wedge s}}{q}\wedge\mathcal{E}}{p \vee q}\vee\mathcal{I}}{r}\supset\mathcal{E}}{q \wedge s \supset r}\supset\mathcal{I},-2}{(p \vee q \supset r) \supset (q \wedge s \supset r)}\supset\mathcal{I},-1$$

2.

$$\dfrac{\dfrac{\overset{+1}{(p \vee q) \wedge \neg p}}{p \vee q}\wedge\mathcal{E} \quad \dfrac{\dfrac{\dfrac{\overset{+1}{(p \vee q) \wedge \neg p}}{\neg p}\wedge\mathcal{E} \quad \overset{+2}{p}}{\bot}\neg\mathcal{E}}{q}\bot\mathcal{E} \quad \overset{+3}{q}}{(p \vee q) \wedge \neg p \supset q}\vee\mathcal{E},-2,-3\supset\mathcal{I},-1$$

3.

$$\dfrac{\dfrac{\dfrac{\dfrac{\overset{+1}{q \supset r} \quad \overset{+2}{q}}{r}\supset\mathcal{E}}{r \vee s}\vee\mathcal{I}}{(r \vee s) \vee \neg q}\vee\mathcal{I} \quad \dfrac{\overset{+3}{\neg q}}{(r \vee s) \vee \neg q}\vee\mathcal{I}}{(r \vee s) \vee \neg q}\text{Dil.},-2,-3}{(q \supset r) \supset (r \vee s) \vee \neg q}\supset\mathcal{I},-1$$

$$\dfrac{\dfrac{\dfrac{\dfrac{\overset{+1}{\neg(p \supset q)} \quad \dfrac{\dfrac{\overset{+3}{\neg p} \quad \overset{+4}{p}}{\bot}\neg\mathcal{E}}{q}\bot\mathcal{E}}{p \supset q}\supset\mathcal{I},-4}{\bot}\neg\mathcal{E}}{p}\bot\mathcal{E} \quad \overset{+2}{p}}{p}\text{Dil.},-2,-3}{\neg(p \supset q) \supset p}\supset\mathcal{I},-1$$